

OVERVIEW

I am a physician-scientist in-training at the University of Pennsylvania. My research focuses on how we can build trustworthy, evidence-based ML systems for healthcare. I leverage [prior knowledge](#) and [statistical methods](#) to adapt algorithms to different patient populations, with applications in [rare disease diagnostics](#), [clinical decision making](#), and [precision medicine](#).

EDUCATION

MD-PhD Bioengineering, University of Pennsylvania 2021 - Present
[NIH F30 NRSA Fellow](#) | HHMI-NIBIB T32 Fellow | Medical Education Concentration
Advised by [Osbert Bastani](#) and [James Gee](#) | [PhD Thesis Link](#) | [Research Award](#)

MS Computer Science, University of Pennsylvania 2023 - 2025
Penn Engineering [Teaching Award](#)

BS Applied Physics, California Institute of Technology 2017 - 2021
Salutatorian | Full-Ride Merit Scholarship Award | Advised by [Mikhail Shapiro](#)

EXPERIENCE

Research Scientist, Abridge New York City, NY | 2026 - Present
Evaluated the trustworthiness and reliability of AI-based clinical decision support tools in clinical workflows. Collaborated with external partners on applications of AI in medical education.

ML Scientist Intern, Genentech San Francisco, CA | 2025
Designed algorithms for optimizing personalized treatment strategies and efficient patient-clinical trial matching using large language models (LLMs).

AI Research Consultant, Scale AI Remote | 2025
Engineered medical reasoning benchmarks for evaluating the clinical capabilities of frontier LLMs. Critiqued and red-teamed medical usage of proprietary LLMs prior to public use.

Health VC Fellow, 25madison New York City, NY | 2024
Bridged clinical and engineering teams to help drive clinical operations for stealth healthtech incubations. Led exploratory investment research into digital health sectors.

AI Clinical Fellow, Glass Health Remote | 2023 - 2024
Authored and assessed clinical guideline articles as knowledge sources for large language models (LLM). Investigated applications of LLMs for medical education.

PhD Research Intern, Microsoft Research Redmond, WA | 2022
Developed ML methods for accelerated MRI imaging. Proposed novel techniques for better generalization of MRI image reconstruction models.

Software Engineer, Hyperfine Guilford, CT | 2021
Developed ML methods for accelerated MRI imaging and improving the generalizability of MRI image reconstruction models to different anatomies and patient populations.

SELECTED WORKS

- [1] **Yao MS**. Large language model-based evaluation of the impact of gender in medical research. **JAMIA Open**. (2026). [Link](#)
- [2] **Yao MS**, Bastani O, Andersson A, Biancalani T, Bentaieb A, Iriondo C. Knowledgeable language models as black-box optimizers for personalized medicine. **ICLR**. (2025). [Link](#)
- [3] **Yao MS**, Chae A, Saraiya P, Kahn CE, Witschey WR, Gee JC, Sagreiya H, Bastani O. Evaluating acute image ordering for real-world patient cases via language alignment with radiological guidelines. **Nat Commun Med**. (2025). [Link](#) | [Penn News](#) | [Research Award](#)
- [4] **Yao MS**, Huang L, Leventhal E, Sun C, Stephen SJ, Liou L. Leveraging datathons to teach AI in undergraduate medical education. **JMIR Medical Education**. (2025). [Link](#)
- [5] **Yao MS**, Gee JC, Bastani O. Diversity by design: Leveraging distribution matching for offline model-based optimization. **ICML**. (2025). [Link](#)
- [6] Wu Y, Liu Y, Yang Y, **Yao MS**, Yang W, Shi X, Yang L, Li D, Liu Y, Gee JC, Yang X, Wei W, Gu S. A concept-based interpretable model for the diagnosis of choroid neoplasias using multimodal data. **Nature Communications**. (2025). [Link](#)
- [7] **Yao MS**, Zeng Y, Bastani H, Gardner J, Gee JC, Bastani O. Generative adversarial model-based optimization via source critic regularization. **NeurIPS**. (2024). [Link](#)
- [8] Yang Y, Gandhi M, Wang Y, Wu Y, **Yao MS**, Callison-Burch C, Gee JC, Yatskar M. A textbook remedy for domain shifts: Knowledge priors for medical image analysis. **NeurIPS (Spotlight)**. (2024). [Link](#)

TEACHING

Instructor and Curriculum Lead, **Ethical Algorithms for the Modern Clinician** | [Link](#)
Head TA, **Pre-Clinical Medicine** (Clerkship Prep, Penn) | 2023, 2026
Head TA, **Healthcare and Technology** (CIS 7000, Penn) | 2023, 2024, 2026
TA, **Distributed Computer Systems** (CIS 5050, Penn) | 2025-2026
TA, **Principles of Deep Learning** (ESE 5460, Penn) | 2024
TA, **Imaging Informatics** (EAS 5850, Penn) | 2024
TA, **Diagnostic Ultrasound for Medical Students** (Penn) | 2023-2024

Head TA, **Applied Mathematics** (ACM 95ab, Caltech) | 2020-2021
TA, **Graduate Classical Physics** (Ph 106a, Caltech) | 2020
TA, **Quantum Physics** (Ph 12ab, Caltech) | 2019-2020
TA, **Operating Systems** (CS 24, Caltech) | 2019
TA, **Special Relativity and Electromagnetism** (Ph 1bc, Caltech) | 2018-2019

SERVICE

Organizations

Co-President, [PennHealthX](#) Student Group | 2025-Present
Co-President, Medical Education Club, University of Pennsylvania | 2026-Present
Technology Committee Vice-Chair, [American Physician Scientists Association](#) | 2023-2025
Director of Data Science & AI, [MDplus](#) | 2023-2025
AI Curriculum Task Force Member, University of Pennsylvania | 2023-2024, 2026-Present
Admissions Committee, University of Pennsylvania | 2023-2024

Editor-in-Chief, [Caltech Undergraduate Research Journal](#) | 2020-2021
Volunteer Tutor, [Caltech RISE Tutoring Program](#) | 2019-2021
Peer Tutor, Caltech Deans' Office | 2018-2021
Student Body Representative, [Caltech Academics and Research Committee](#) | 2017-2019

Referee

AMIA Annual Symposium

npj Digital Medicine

Neural Information Processing Systems (NeurIPS)

International Conference on Learning Representations (ICLR)

International Conference on Machine Learning (ICML)

AAAI Conference on Artificial Intelligence

AHLI Conference on Health, Inference, and Learning (CHIL)

RSNA Radiology: AI (Editor's Recognition Awardee - [Reviewer with Distinction](#))

Invited Talks

[Digi-Stars Digital Health Symposium](#), University of Utah | 2026

[Fueling Reimagination to Advance Medical Education](#), University of Pennsylvania | 2026

ML for Clinical Medicine and Education, Tufts University | 2026

INVITED EDITORIALS

[9] **Yao MS**, Chae A. Refining the second reader: Artificial intelligence as a triage partner in population-based breast cancer screening. **Radiology: AI**. (2026). [Link](#)

[10] Gee JC, **Yao MS**. Effective structured information extraction from chest radiography reports using open-weights large language models. **Radiology**. (2025). [Link](#)

TEXTBOOK CHAPTERS

[11] **Yao MS**, Johnson KB. Generative AI in healthcare. In **Health, Healthcare and Technology** (Eds. Johnson KB, Merchant R). Springer Nature.

PREPRINTS

[12] Leventhal E, Suresh S, Rajan A, Bunch K, Kunisetty B, Roytman MV, Ipe J, Mallahan S, **Yao MS**. Improving AI readiness among medical trainees through an interdisciplinary health datathon: Mixed methods evaluation. Preprint. (2026). [Link](#)

ADDITIONAL WORKS

[13] Barinaga Z, Chae A, Hashimoto DA, Jiang J, Leventhal E, Meng S, Purcell M, **Yao MS**. Ethical algorithms for the modern clinician. **NeurIPS Education Program**. (2025). [Link](#)

[14] Chae A†, **Yao MS**†, Sagreiya H, Goldberg AD, Chatterjee N, MacLean MT, Duda J, Elahi A, Borthakur A, Ritchie MD, Rader D, Kahn CE, Witschey WR, Gee JC. Strategies for implementing ML algorithms in the clinical practice of radiology. **Radiology**. (2024). [Link](#)

[15] **Yao MS**†, Chae A†, MacLean MT, Verma A, Duda J, Gee JC, Torigian DA, Rader D, Kahn CE, Witschey WR, Sagreiya H. SynthA1c: Towards clinically interpretable patient representations for diabetes risk stratification. **Prime MICCAI**. (2023). [Link](#)

[16] **Yao MS**, Hansen MS. A path towards clinical adaptation of accelerated MRI. **Machine Learning for Health**. (2022). [Link](#)

[17] Abedi MH†, **Yao MS**†, Mittelstein DR, Bar-Zion A, Swift MB, Lee-Gosselin A, Barturen-Larrea P, Buss MT, Shapiro MG. Ultrasound-controllable engineered bacteria for cancer immunotherapy. **Nature Communications**. (2022). [Link](#) | [Patent](#)

[18] **Yao MS**, Uhr L, Daghlain G, Amrute JM, Deshpande R, Mathews B, Patel SA, Henri R, Liu G, Reiersen K, Johnson G. Demonstration of a longitudinal medical education model to teach point-of-care ultrasound in resource-limited settings. **POCUS J**. (2020). [Link](#)

PERSONAL PROJECTS

Medical Education: [Clerkship Connections](#) | [Pimping Rounds](#)

Programming Puzzles: [Advent of Code 2025](#) | [Project Euler Puzzles](#)